

CONFIDENTIAL — PROJECT BLUEPRINT

PREPARED 27 JULY 2026 · V1.0

SkillChain

A learning platform built to hold a real business, not just a course catalog.

This is the complete blueprint for an enterprise-grade course marketplace and learning experience platform: what it does, how it earns, how it's built, and the plan to ship it. Every figure on this page is pulled from architecture, data, and API documentation that already exists — not projections.

24

backend service domains

85

normalized data tables

125

documented API routes

11

distinct user roles

13

delivery epics scoped

WHY THIS, WHY NOW

Off-the-shelf course platforms weren't built to be owned

Course marketplaces and cohort-based learning have converged: students expect self-paced video, live classes, and AI-assisted study in one place; instructors expect real payout infrastructure, not a PayPal link. Building on someone else's platform means renting your own customer relationship and margin.

SkillChain is architected as a platform a business can actually run — **five payment rails including M-Pesa, Flutterwave, and Paystack** for markets most competitors ignore, an

instructor revenue-share and payout engine, and an affiliate program, all as first-class parts of the data model rather than bolted-on plugins.

WHAT WE'RE BUILDING

Six capability areas, engineered as one coherent system

LEARNING

Self-paced and live, in one player

Course playback sits alongside instructor-hosted live classes over Zoom or Google Meet — same enrollment, same progress record.

- Resumable video with notes, bookmarks, transcripts
- Live session scheduling, registration, and gated join links
- QR-verifiable completion certificates

ASSESSMENT

Mastery you can prove

Three assessment types cover knowledge checks through to applied skill.

- Quizzes with attempt limits and pass thresholds
- Instructor-graded assignments
- Judged coding exercises with hidden test cases

COMMERCE

Every way people actually pay

One-time purchase, subscriptions, bundles, and regional payment methods, all reconciled to a single ledger.

- Coupons, promotions, and gift cards
- Subscription plans with proration and cancellation
- Instructor payouts and affiliate commissions

AI

Assistance built into the workflow

Eleven AI capabilities, each tied to a real moment in the student or instructor journey.

- Course-scoped tutor chat and recommendations
- Auto-generated summaries, flashcards, quizzes
- Transcript, subtitle, and grading assistance

INSTRUCTOR

A dashboard that runs a business

Authoring is one part of the job — the rest is revenue, engagement, and scheduling.

- Course authoring and content review workflow
- Revenue, engagement, and drop-off analytics
- Live session hosting and coupon management

GOVERNANCE

Trust and control, by design

Eleven distinct roles and an immutable trail on everything that touches money or access.

- Role-scoped permissions from Guest to Super Admin
- Content moderation and fraud/refund risk checks
- Immutable audit log on privileged actions

WHO IT SERVES

Eleven roles, one platform

Every workflow below is scoped to a specific role with its own permissions — nothing is a shared "admin does everything" catch-all.

Guest Browses and previews the catalog before creating an account.	Student Enrolls, learns, assesses, and earns certificates.	Instructor Authors courses, hosts live sessions, earns revenue.
Organization Manages workforce learning across a team or company.	Affiliate Refers students and earns commission on conversions.	Moderator Reviews flagged content and community conduct.
Support Agent Owens the support ticket queue and SLA.	Finance Officer Reconciles payments, payouts, and refunds.	Content Reviewer Approves or rejects courses before publication.

Administrator

Operates day-to-day platform configuration.

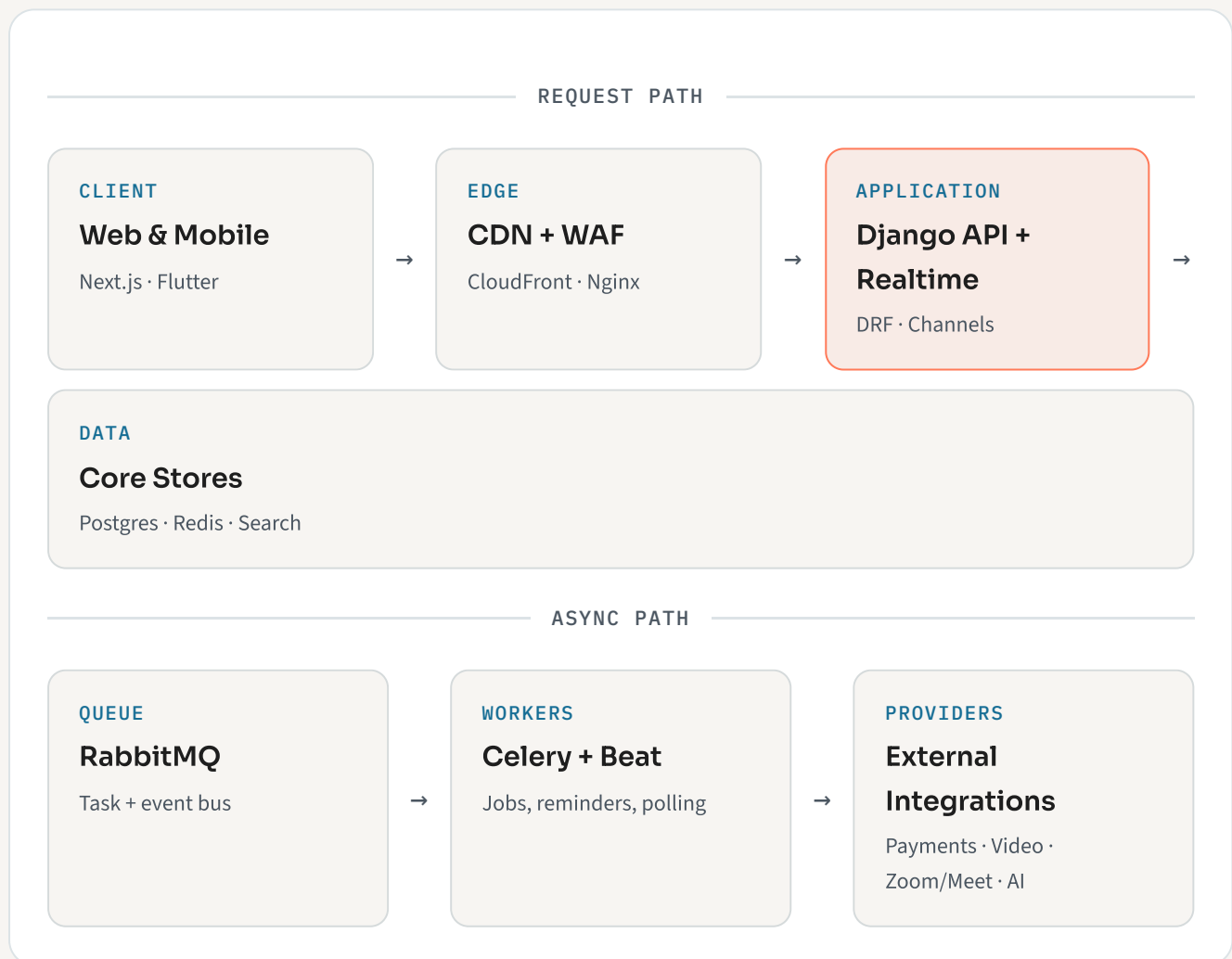
Super Administrator

Holds platform-wide, irreversible-action authority.

HOW IT'S BUILT

A modular monolith today, built to split tomorrow

The backend ships as one deployable system organized into 24 clearly bounded domains — fast to build and reason about now, with the seams already in place to extract search, notifications, payments, or recommendations into independent services the moment traffic demands it.



Django

DRF

PostgreSQL

Redis

Next.js

Flutter

Docker

Celery

RabbitMQ

Channels

Kubernetes

AWS

GitHub Actions

Elasticsearch

PAYMENTS

Stripe

PayPal

Flutterwave

Paystack

M-Pesa

VIDEO, LIVE & AUTH

Mux

Bunny

Vimeo

Zoom

Google Meet

Google OAuth

Apple OAuth

SECURITY & RELIABILITY

Built to be trusted with money and student data

✓ Rotating JWT + optional MFA

15-minute access tokens, 30-day rotating refresh tokens with reuse detection.

✓ Encryption everywhere it matters

KMS-managed encryption at rest, TLS 1.2+ in transit, encrypted OAuth tokens.

✓ OWASP-aligned controls

CSRF, XSS, and SQL-injection mitigations built into the framework layer, not bolted on.

✓ Immutable audit trail

Every privileged and financial action is logged and cannot be altered after the fact.

✓ Protected video & live sessions

Signed, watermarked playback; live join links gated to registered, enrolled students only.

✓ 99.95% / 99.99% availability targets

Platform-wide and checkout SLOs, with a 2-hour RTO and 15-minute RPO disaster recovery plan.

DELIVERY ROADMAP

From foundations to global scale in four quarters

Q1	Q2	Q3	Q4
Marketplace core	Enterprise & scale	AI & localization	Global resilience
Catalog, checkout, learning player, and instructor authoring — the sellable MVP.	Organization controls, advanced analytics, affiliate program maturity.	AI tutor and recommendation maturity, deeper regional payment and language support.	Multi-region optimization and ecosystem integrations.

Q1 breaks down into 8 two-week sprints, each ending in a working increment — foundations and CI/CD first, hardening and launch readiness last.

WHAT'S ALREADY DONE

This isn't a pitch — it's a documented system

Before a line of application code is written, the platform has been fully specified: 13 documents covering product intent, architecture, data, and delivery — including 41 formal functional requirements and 44 scoped user stories across 13 epics.

- 00

Product

✓ PRD — vision, KPIs, monetization model, release strategy

✓ SRS — 41 functional and non-functional requirements

- 01

Architecture

✓ Technical architecture — components, scaling, evolution plan

✓ Infrastructure diagram — full AWS/Kubernetes topology

- 02

Database

✓ ERD and normalized schema — 85 tables across 10 domains

- 03

API

✓ API documentation — 125 documented routes across 17 domains

✓ OpenAPI 3.1 specification, validated

04

Platform Structure

- ✓ Monorepo layout, 24 Django apps, Next.js and Flutter structure
- ✓ Core business flows — auth, enrollment, payment, live sessions, certification

05

Product Design

- ✓ UX journeys, wireframes, component library, design system tokens

06

DevOps, Security & QA

- ✓ CI/CD, Docker, Kubernetes, monitoring, backup, and DR plans
- ✓ Security architecture, test plan, and QA checklist

07

Delivery Planning

- ✓ 13 epics, 44 user stories, sprint plan, and quarterly roadmap

NEXT STEPS

From blueprint to first sprint

01

Review the blueprint

Walk through this document and the underlying 13-doc set together.

02

Confirm scope & priorities

Lock the Q1 feature set and any market-specific priorities (payment rails, languages).

03

Kick off Sprint 1

Auth core, repository, CI/CD baseline, and environments — foundations first.

04

First milestone demo

Working catalog and checkout flow at the end of the Q1 sprint cycle.

Ready to move from documentation to a working platform.

SkillChain Engineering

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SkillChain Platform Blueprint

Confidential — prepared for internal and client review